Over the past year, CB M&S researchers have conducted numerous Institutional Review Board-approved real-time experiments. This poster will highlight FY04/1QFY05 efforts, and give a colorful overview of the present and future of the CB M&S program. Distinct information will be provided covering the experiments themselves, collaboration with organizations such as DMSO and HECOE, as well as the latest on the procedures used to transfer real-time behavioral data into computer modeling and simulation. Emergent Hierarchies in Perception Thomas Malloy, Psychology, University of Utah Mathematical models have formed a rigorous basis for defining emergence within the logic of the models. More problematic has been the mapping of mathematically defined emergence onto scientific phenomena. Bateson's difference-based epistemology can be simulated by a Boolean network model. Bateson proposed that taking differences in differences would produce emergent hierarchies of knowledge. Simulating this proposal, the Boolean model, by taking differences in differences, generates hierarchical categories of perceptual stimuli. The crucial result is that these model-generated categories correspond to human perceptual categorical judgments. Thus the model- defined emergence hierarchy maps to perceptual hierarchies. Three realms of description will converge on the concept of emergence: Batesonian epistemology, a Boolean model, and human perceptual experience. In this case at least, this convergence grounds model-defined emergence both to a broad epistemological framework and to human experience. Resemblance in Morphogenesis Thomas Malloy, Psychology, University of Utah The resemblance of one form to another is a phenomenon of basic interest in perception. In a Boolean net model attractor cycles are coded as a sequence of state vectors (of 0 s and 1 s) easily expressible as visual forms. This expression corresponds to the idea of morphogenesis. Let the forms resulting from two attractor cycles be Parent 1 and Parent 2. Take two state vectors, one from each parent, and create a Child vector by splicing together half the 0 s and 1 s from one parent and half the 0 s and 1 s from the other to create a new state vector, analogous to gene splicing. Run the system starting from the new state vector. Under very broad constraints, the system will fall into a third, Child, attractor cycle, which expressed as a visual form will resemble both Parent 1 and Parent 2 while having its own unique form. The Myth of Psyche Through Nonlinear Eyes Terry Marks-Tarlow, Clinical Psychologist, Private Practice, CA, Training Faculty, Southern California Counseling Center An aspect of nonlinear science in context is how new ideas fit into old systems. One context vital to the pursuit of meaning in psychology is ancient mythology. With historical precedents set by Freud, Jung, May, and Hillman, clinical practice is frequently guided by originating myths that supply root metaphors and narratives. This paper analyzes the myth of Psyche, derivation of our field s name psychology and its subject matter the psyche. Nonlinear wisdom emerges through several important themes: the necessity of living in the dark with chaos; the self-organizing quality of nature, both inner and outer; interpenetrating, fractal boundaries between self, world, and other; and the paradoxical essence of psyche, where only through separation and individuation can we open ourselves to love and vital interconnection with others. Controlling Duopoly Chaos Akio Matsumoto, Economics, Chuo University The recently developing theory of nonlinear dynamics reveals that any economic model can generate complex dynamics involving chaos if its nonlinearities become strong enough. The main purpose of this study is to consider economic implications of generating chaos as well as controlling chaos in a Cournot duopoly model with unimodal reaction functions. An equivocal characteristic of economic chaos is demonstrated. From the long-run point of view, one of the duopolists can be beneficial, and the other harmful in the chaotic market in a sense that the long-run average profit taken along chaotic trajectories is larger than the one of the latter. Then this study applies two distinctive control methods, the adaptive control method and the pole-placement method to the chaotic markets, in order to show that putting control reverses the situation: the beneficial duopolist in the chaotic market becomes disadvantageous and the harmful duopolist advantageous in the controlled market. This implies that either way of generating chaos or controlling chaos is unable to make both duopolists happy together. 